

Use Case

I want a system that can automatically scan my entire database, find data-quality issues like inconsistent values or placeholders, fix what can be fixed with SQL, and create a clean report that's shared via email.

No more manual checks, and no more missed cleanup.

Mindset / Thought Process

1. I want to keep my database clean and audit-ready, without needing to go table-by-table manually.
 2. Some issues are simple (like wrong date formats) and can be fixed directly with SQL.
 3. Others need team discussion, so I want to classify them accordingly.
 4. I also want full visibility of what was found, what was fixed, and what needs attention.
 5. And I want all of this to be documented and shared automatically with my team.
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Solution Design

- A button I can click to run the audit whenever I want.
 - An AI agent that scans the entire database for data-quality problems.
 - A system to separate issues that can be fixed from those needing policy decisions.
 - A process that asks for approval before applying SQL fixes.
 - Another AI agent that writes and runs SQL queries for each fixable issue.
 - A final agent that creates a clean, well-formatted Google Doc with the full report.
 - Email delivery with a link to the final report.
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Step-by-Step Execution

Step 1: Start the Workflow

- Use the **Manual Trigger** node in n8n.
 - You can later replace this with a scheduled trigger (e.g., weekly or monthly check).
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Step 2: Scan the Database for Issues

- The **Data Health Checker Agent** reads the full schema and samples data from your PostgreSQL database.
 - It identifies issues like:
 - Placeholder values ("N/A", "UNKNOWN")
 - Inconsistent formats (e.g., mixed date styles)
 - Orphaned keys, NULLs in required fields, etc.
 - The agent returns a list of issues in a structured JSON format.
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Step 3: Filter Issues That Can Be Auto-Fixed

- Use a **Filter Node** to check if the issue is marked as “Fixable with SQL.”
 - These issues are routed to the auto-fix flow. Others are logged for later review.
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Step 4: Get Manager’s Approval

- Send an email to the stakeholder asking:
“Are you available to review and approve SQL fixes?”
 - Once they respond YES, continue. Otherwise, stop the run.
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Step 5: Send Individual Approvals for Each Issue

- Loop through each issue one by one.
- For each, send a detailed email with:

- Description of the issue
 - Proposed fix
 - Button to approve or reject
 - Wait for response before proceeding to fix.
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Step 6: Fix the Data Automatically

- If approved, the **Data Issue Fixer Agent**:
 - Generates the appropriate SQL
 - Runs the SQL directly on the DB using a connected tool
 - Returns whether the fix was successful, and the exact query used
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Step 7: Merge Fix Results

- Merge the original issue list with the SQL fix outputs
 - You now have a complete picture: what was fixed, what failed, and what wasn't touched
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Step 8: Create the Report

- The **Data Health Reporter Agent**:
 - Groups the fixed issues and pending ones
 - Creates a professional Google Doc with all details
 - Formats it neatly with headings, spacing, and bullet points
 - Shares it via Google Drive with public “view” access
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Step 9: Deliver the Report

- Send a final email with the link to the generated report.

- Now your team knows what was found, what was fixed, and what needs discussion.
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Final Outcome

- I run the audit with a click
- The system checks and fixes what it can, only after I approve
- It sends me a clean report every time
- My database stays healthy, my team stays informed
- And I don't have to waste time scanning tables manually anymore